

Differential Gene Expression Analysis

Effect of Ethanol Exposure on the Adult Mouse Brain Transcriptome

Dataset	GSE222445 (NCBI Gene Expression Omnibus)
Organism	Mus musculus (adult mouse brain)
Platform	Google Colab R 4.5.3 Bioconductor 3.22
Tools	DESeq2 · apeglm · GEOquery · clusterProfiler · ReactomePA
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Transcriptomic Analysis of Ethanol-Induced Neuroinflammation

Objective

The main question I wanted to answer is: what happens to the brain's genes when someone drinks heavily over a long period? For this project, I looked at how chronic alcohol exposure changes gene expression in the cerebellum (a part of the brain). My goal was to understand the biology behind brain inflammation caused by alcohol toxicity.

Dataset

I used a public dataset (named GSE222445) containing RNA sequencing data from the brains of adult mice. The data includes 10 samples in total: 5 from mice that were given alcohol (the 'Ethanol' group) and 5 from healthy mice (the 'Control' group). By comparing these two groups, I can see exactly which genes are turned on or off by the alcohol.

Results

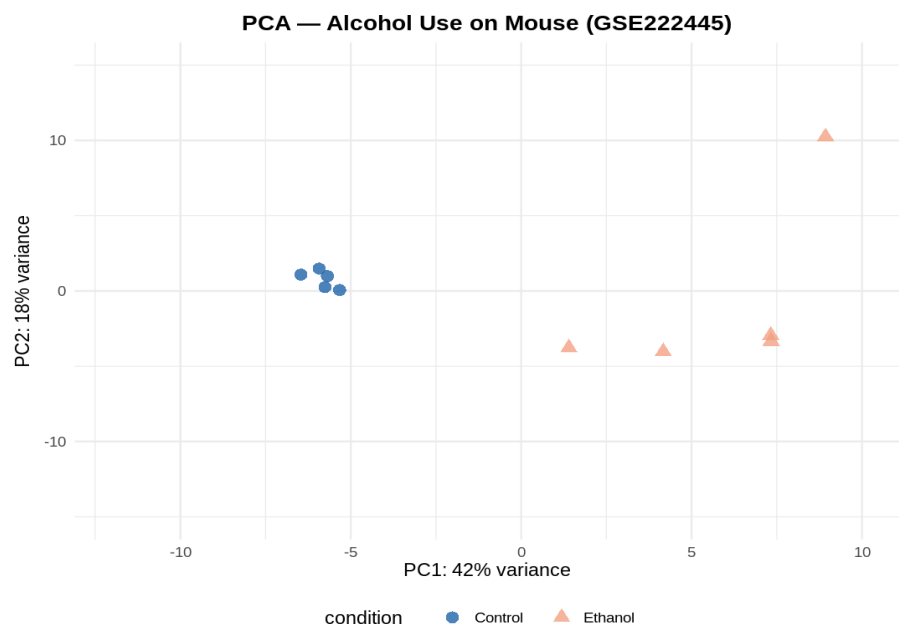
I used a tool called DESeq2 to analyze nearly 24,000 genes. Here is what I found:

- **Significant Changes:** I found 141 genes that were significantly altered by the alcohol exposure.
- **Ups and Downs:** Out of those 141 genes, 86 became more active (upregulated) and 55 became less active (downregulated) compared to the healthy mice.
- **Top Gene:** The gene that changed the most was *Gpx3*. This gene is normally responsible for fighting off cell damage and oxidative stress.
- **Pathways:** When I looked at what these changed genes actually do, I found they are mostly involved in the body's immune response, like white blood cell movement and healing processes.

Plot Analysis

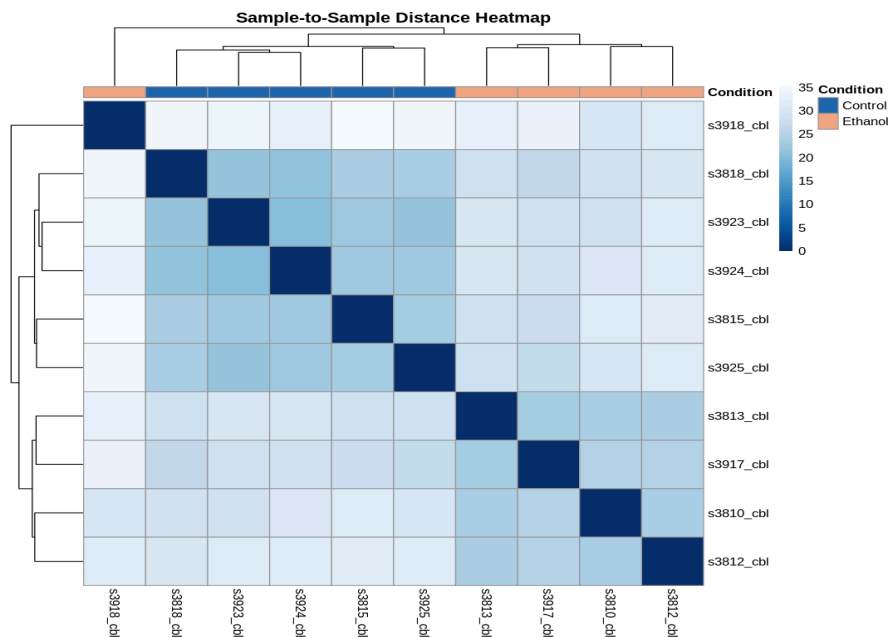
To truly understand the data, I generated several visual plots to track the quality of my samples, identify significant gene changes, and map out the biological pathways affected.

1. PCA Plot (Sample Quality)



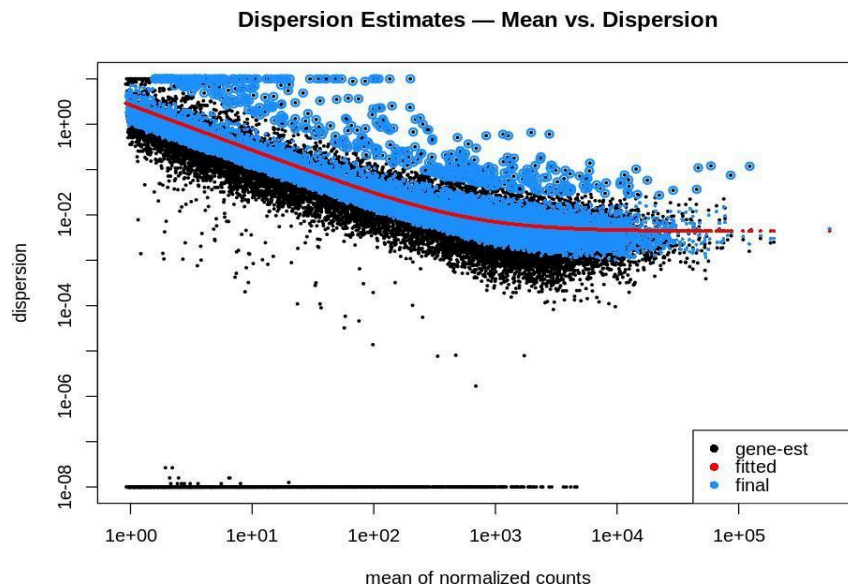
The PCA plot checks if my samples group together logically based on their overall gene expression. I can see a clear separation between the Ethanol and Control groups. This is a great sign—it means the alcohol treatment had a strong, distinct effect on the brain's gene activity that outweighs random biological noise.

2. Sample Distance Heatmap



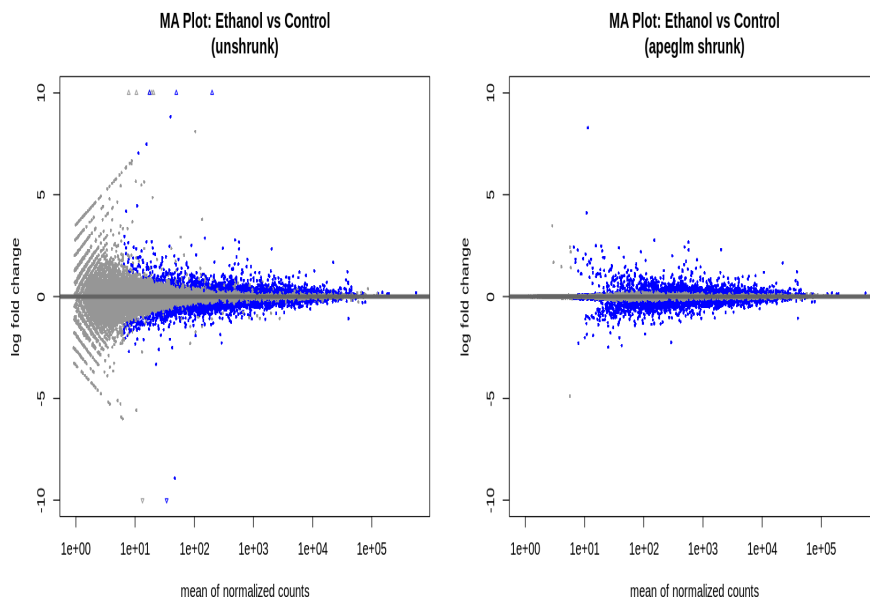
Similar to the PCA, this heatmap visualizes the distance (or similarity) between each individual sample. Darker colors mean the samples are highly similar. Again, I can see the samples neatly clustering by their treatment group, confirming the quality of my data and the absence of experimental errors.

3. Dispersion Plot



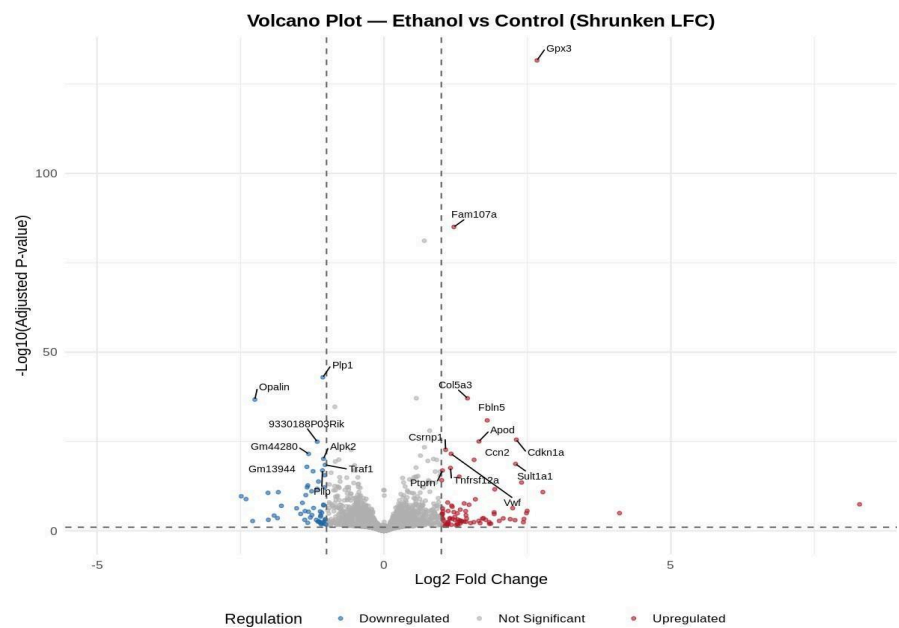
This plot shows how the variance in gene expression changes with the average expression level, verifying that my statistical model fits the data properly.

4. MA Plot (Fold Change vs. Average Expression)



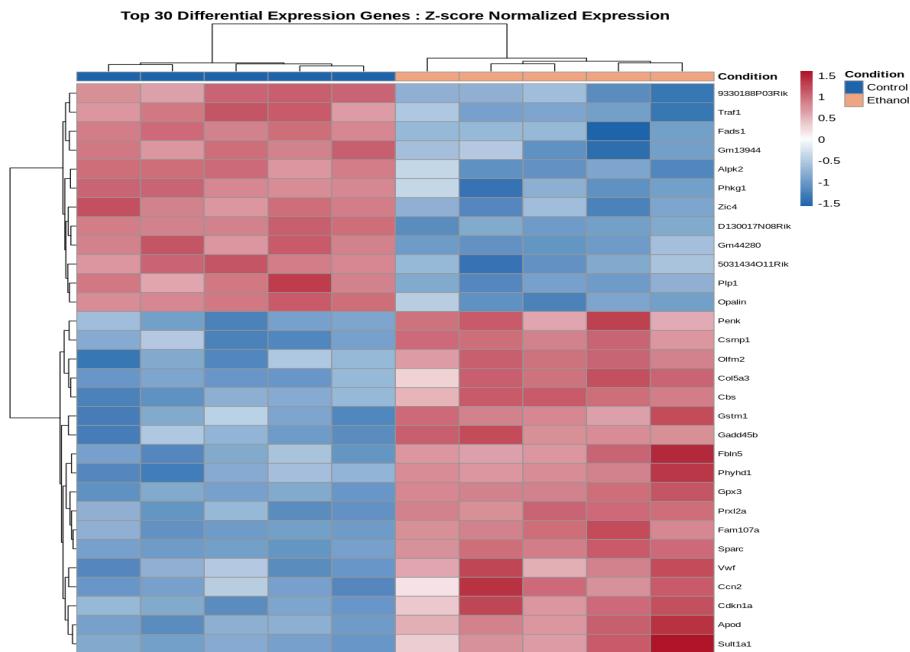
The MA plot maps the fold change of genes (how much they went up or down) against their average expression level. I used a statistical technique (shown in the 'shrunk' plot on the right) to reduce the noise from genes with very low expression. The points scattered far from the center zero-line are my true, confident discoveries.

5. Volcano Plot (Statistical Significance)



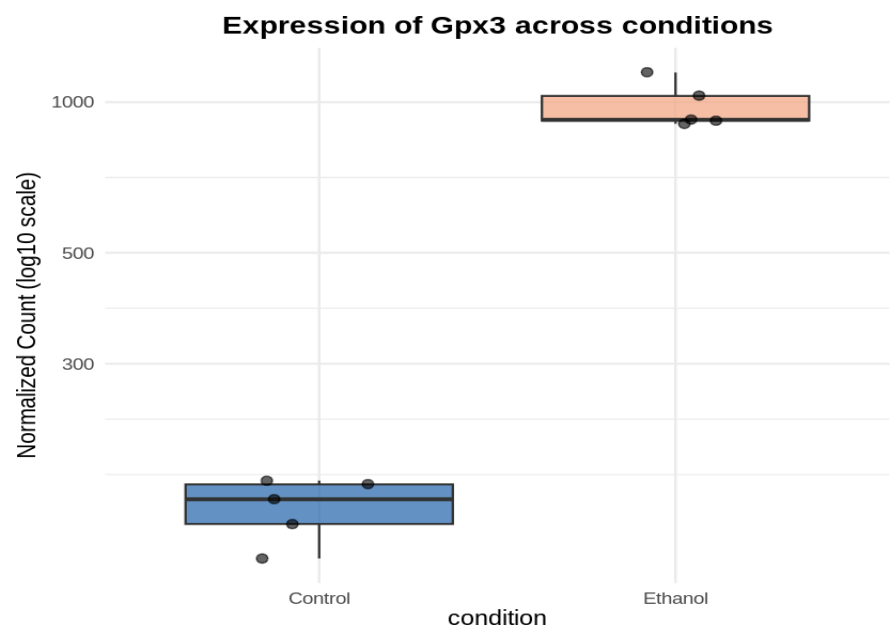
The volcano plot combines statistical significance (y-axis) with the magnitude of change (x-axis). The genes pushed into the upper corners are my most confident hits. The dots mapped on the far edges represent genes that were significantly upregulated or downregulated by alcohol.

6. Top 30 Genes Heatmap



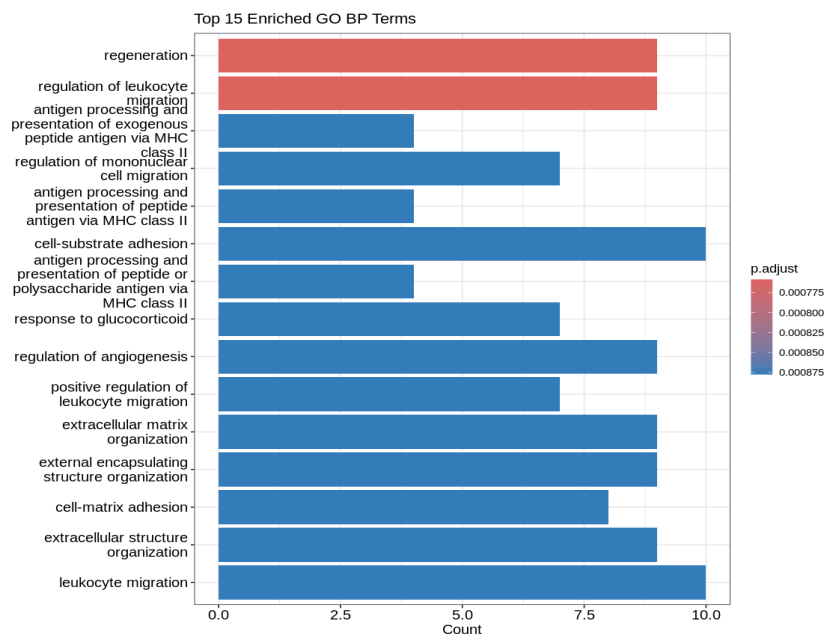
This heatmap shows top 30 most significantly altered genes across all samples. It clearly shows a stark contrast: genes that are 'hot' in the ethanol group are 'cold' in the control group, and vice versa. This confirms a consistent biological response across the different mice.

7. Top Gene Expression (Gpx3)



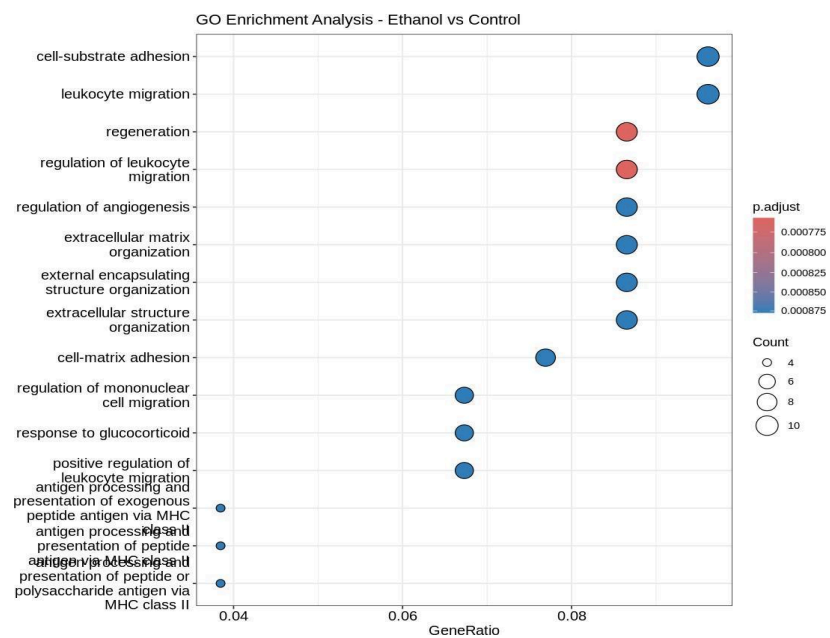
Now, my absolute top candidate gene: Gpx3. The boxplot explicitly compares its normalized read counts between the two groups. Gpx3 is heavily involved in fighting oxidative stress, and its massive spike highlights the tissue's attempt to combat cellular damage.

8. Gene Ontology (Bar Plot)



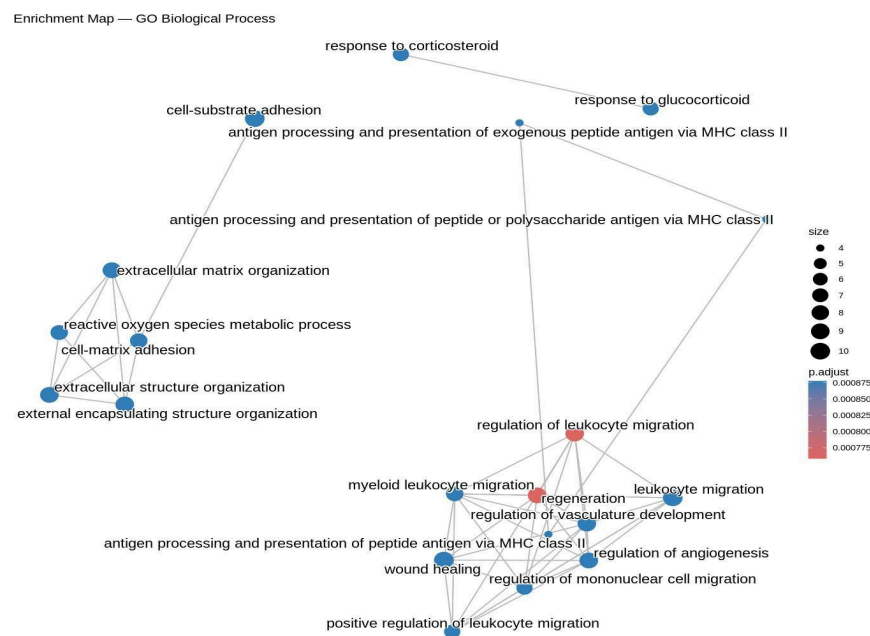
I grouped my altered genes to see what biological pathways they control. The bar plot shows that the most affected pathways are related to immune responses—specifically 'regulation of leukocyte migration' and 'antigen presentation'.

9. Gene Ontology (Dot Plot)



The dot plot further confirms the pathway enrichment, proving the brain is undergoing severe inflammation.

10. Enrichment Map



This map connects the GO terms with shared gene memberships, highlighting the network of biological processes responding to the alcohol exposure.

Annexure — Analysis Code

```
pkg_libs <- paste(
  "libcurl4-openssl-dev libxml2-dev libssl-dev",
  "libfontconfig1-dev libharfbuzz-dev libfribidi-dev",
  "libpng-dev libtiff5-dev libjpeg-dev"
)
system(paste("apt-get install -y -qq", pkg_libs, "2>/dev/null"))
message("System-level dependencies are ready.")

setup_bioc <- function(pkgs) {
  BiocManager::install(pkgs, update = FALSE, ask = FALSE)
}

if (!requireNamespace("BiocManager", quietly = TRUE))
  install.packages("BiocManager", repos = "https://cloud.r-project.org")

setup_bioc("DESeq2")
setup_bioc("apeglm")
setup_bioc("GEOquery")

cran_pkgs <- c("pheatmap", "ggplot2", "ggrepel", "RColorBrewer")
install.packages(cran_pkgs, repos = "https://cloud.r-project.org", quiet = TRUE)

setup_bioc(c("org.Hs.eg.db", "clusterProfiler", "enrichplot"))
setup_bioc("ReactomePA")
message("\nAll R packages successfully installed.")

if (!requireNamespace("org.Mm.eg.db", quietly = TRUE))
  BiocManager::install("org.Mm.eg.db", update = FALSE, ask = FALSE)
library(org.Mm.eg.db)
message("Mouse annotation database (org.Mm.eg.db) loaded.")

pkgs_to_load <- c(
  "DESeq2", "apeglm", "GEOquery",
  "pheatmap", "ggplot2", "ggrepel", "RColorBrewer",
  "org.Mm.eg.db", "clusterProfiler", "enrichplot", "ReactomePA"
)
invisible(lapply(pkgs_to_load, library, character.only = TRUE))
message("\nAll libraries loaded.")
cat("DESeq2 :", as.character(packageVersion("DESeq2")), "\n")
cat("R :", R.version.string, "\n")

system2("gunzip", "GSE222445_manuscript.adults.raw.counts.csv.gz")
raw_df <- read.csv("GSE222445_manuscript.adults.raw.counts.csv",
  header = TRUE, check.names = FALSE)
gene_ids <- raw_df[, 1]

head(raw_df)
counts_mat <- as.data.frame(raw_df[, -1])
rownames(counts_mat) <- make.unique(gene_ids)
head(counts_mat)

gse_obj <- getGEO("GSE222445")
pheno <- pData(gse_obj[[1]])
sample_info <- data.frame(
  condition = factor(pheno$treatment:chl, levels = c("Control", "Ethanol")),
  rownames = pheno$title
)
sample_info
sample_info <- sample_info[order(sample_info$condition), , drop = FALSE]
```

```

counts_mat <- counts_mat[, sample_info$rownames]
counts_mat <- counts_mat[order(rownames(counts_mat)), ]
head(counts_mat)
sample_info

dds_obj <- DESeqDataSetFromMatrix(
  countData = as.matrix(round(counts_mat)),
  colData = sample_info,
  design = ~ condition
)
dds_obj
row_totals <- rowSums(counts(dds_obj))
dds_obj <- dds_obj[row_totals >= 10, ]
dds_obj

vst_obj <- vst(dds_obj, blind = TRUE)
message("VST complete.")
pca_df <- plotPCA(vst_obj, intgroup = "condition", returnData = TRUE)
pct_var <- round(100 * attr(pca_df, "percentVar"))
grp_colors <- c("Control" = "#2166AC", "Ethanol" = "#F4A582")
options(repr.plot.width = 9, repr.plot.height = 7)
p_pca <- ggplot(pca_df, aes(PC1, PC2, color = condition, shape = condition)) +
  geom_point(size = 4, alpha = 0.8) +
  scale_color_manual(values = grp_colors) +
  xlim(c(-12, 10)) + ylim(c(-15, 15)) +
  xlab(paste0("PC1: ", pct_var[1], "% variance")) +
  ylab(paste0("PC2: ", pct_var[2], "% variance")) +
  ggtitle("PCA \u2014 Alcohol Use on Mouse (GSE222445)") +
  theme_minimal(base_size = 14) +
  theme(legend.position = "bottom",
        plot.title = element_text(hjust = 0.5, face = "bold"))
print(p_pca)

```

```

dist_mat <- as.matrix(dist(t(assay(vst_obj))))
annot_df <- data.frame(Condition = sample_info$condition)
rownames(annot_df) <- colnames(counts_mat)
annot_colors <- list(Condition = c("Control" = "#2166AC", "Ethanol" = "#F4A582"))
blue_pal <- colorRampPalette(rev(brewer.pal(9, "Blues")))(255)
options(repr.plot.width = 9, repr.plot.height = 8)
pheatmap(
  dist_mat,
  clustering_distance_rows = dist(t(assay(vst_obj))),
  clustering_distance_cols = dist(t(assay(vst_obj))),
  annotation_col = annot_df,
  annotation_colors = annot_colors, col
  = blue_pal,
main = "Sample-to-Sample Distance Heatmap"
)

dds_obj <- DESeq(dds_obj)
message("\nDESeq2 model fitting done.")
options(repr.plot.width = 8, repr.plot.height = 6)
plotDispEsts(dds_obj, main = "Dispersion Estimates \u2014 Mean vs. Dispersion")

resultsNames(dds_obj)
de_res <- results(
  dds_obj,
  contrast = c("condition", "Ethanol", "Control"), alpha
  = 0.1
)
cat("\n Summary: Ethanol vs Control\n")

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summary(de_res)

cat("Available coefficients:\n")
resultsNames(dds_obj)
shrunk_res <- lfcShrink(
  dds_obj,
  coef = "condition_Ethanol_vs_Control", type
  = "apeglm"
)
cat(" Shrunk LFC: Ethanol vs Control\n")
summary(shrunk_res)

options(repr.plot.width = 14, repr.plot.height = 6)
par(mfrow = c(1, 2))
plotMA(de_res, main = "MA Plot: Ethanol vs Control\n(unshrunk)", ylim = c(-10, 10))
plotMA(shrunk_res, main = "MA Plot: Ethanol vs Control\n(apeglm shrunk)", ylim = c(-10, 10))

par(mfrow = c(1, 1))

vlc_df <- as.data.frame(shrunk_res)
vlc_df$gene <- rownames(vlc_df)
vlc_df$status <- with(vlc_df, ifelse(
  !is.na(padj) & padj < 0.1 & abs(log2FoldChange) > 1,
  ifelse(log2FoldChange > 1, "Up", "Down"), "NS"))
label_set <- vlc_df[order(vlc_df$padj), ]
label_set <- head(label_set[label_set$status != "NS", ], 20)
options(repr.plot.width = 10, repr.plot.height = 8)
p_volcano <- ggplot(vlc_df, aes(log2FoldChange, -log10(padj))) +
  geom_point(aes(color = status), size = 1.2, alpha = 0.6) +
  scale_color_manual(
    values = c("Up" = "#B2182B", "Down" = "#2166AC", "NS" = "grey70"),
    name = "Regulation",
    labels = c("Downregulated", "Not Significant", "Upregulated")) +
  geom_vline(xintercept = c(-1, 1), linetype = "dashed", color = "grey40") +
  geom_hline(yintercept = -log10(0.1), linetype = "dashed", color = "grey40") +
  geom_text_repel(data = label_set, aes(label = gene),
    size = 3, max.overlaps = 20, box.padding = 0.5) +
  labs(x = "Log2 Fold Change", y = "-Log10(Adjusted P-value)",
    title = "Volcano Plot \u2014 Ethanol vs Control (Shrunk LFC)") +
  theme_minimal(base_size = 13) +
  theme(legend.position = "bottom",
    plot.title = element_text(hjust = 0.5, face = "bold"))
print(p_volcano)

top30_genes <- head(rownames(shrunk_res[order(shrunk_res$padj), ]), 30)
expr_mat <- assay(vst_obj)[top30_genes, ]
expr_scaled <- t(scale(t(expr_mat)))
div_pal <- colorRampPalette(c("#2166AC", "white", "#B2182B"))(100)
options(repr.plot.width = 10, repr.plot.height = 10)
pheatmap(
  expr_scaled,
  annotation_col = annot_df,
  annotation_colors = annot_colors,
  cluster_rows = TRUE, cluster_cols = TRUE,
  show_rownames = TRUE, show_colnames = FALSE,
  fontsize_row = 8, color = div_pal,
  main = "Top 30 Differential Expression Genes : Z-score Normalized Expression"
)

best_gene <- rownames(shrunk_res[order(shrunk_res$padj), ])[1]
cat("Most significant gene:", best_gene, "\n")
cnt_df <- plotCounts(dds_obj, gene = best_gene,

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        intgroup = "condition", returnData = TRUE)
options(repr.plot.width = 7, repr.plot.height = 6)
ggplot(cnt_df, aes(condition, count, fill = condition)) +
  geom_boxplot(alpha = 0.7, outlier.shape = NA) +
  geom_jitter(width = 0.2, size = 2.5, alpha = 0.6) +
  scale_fill_manual(values = c("Control" = "#2166AC", "Ethanol" = "#F4A582")) +
  scale_y_log10() +
  labs(y = "Normalized Count (log10 scale)",
       title = paste0("Expression of ", best_gene, " across conditions")) +
  theme_minimal(base_size = 14) +
  theme(legend.position = "none",
        plot.title = element_text(hjust = 0.5, face = "bold"))

de_hits <- shrunk_res[
  !is.na(shrunk_res$padj) & shrunk_res$padj < 0.1 &
  abs(shrunk_res$log2FoldChange) > 1, ]
up_set <- rownames(de_hits[de_hits$log2FoldChange > 0, ])
down_set <- rownames(de_hits[de_hits$log2FoldChange < 0, ])
cat("Significant DE genes (|LFC| > 1, padj < 0.1):\n")
cat(" Upregulated in Ethanol: ", length(up_set), "\n")
cat(" Downregulated in Ethanol:", length(down_set), "\n")
all_de <- c(up_set, down_set)

go_res <- enrichGO(
  gene = all_de, OrgDb = org.Mm.eg.db, keyType = "SYMBOL",
  ont = "BP", pAdjustMethod = "BH",
  pvalueCutoff = 0.1, qvalueCutoff = 0.2, readable = TRUE)
cat("\nTop 10 enriched GO Biological Process terms (Mouse):\n\n")
print(head(go_res, 10))

if (!is.null(go_res) && nrow(as.data.frame(go_res)) > 0) {
  cat("Total enriched GO terms found:", nrow(as.data.frame(go_res)), "\n")
  options(repr.plot.width = 10, repr.plot.height = 8)
  print(barplot(go_res, showCategory = 15, title = "Top 15 Enriched GO BP Terms"))
  print(dotplot(go_res, showCategory = 15, title = "GO Enrichment Analysis - Ethanol vs Control"))
} else {
  cat("No enriched GO terms found with current thresholds.\n")
}

head(as.data.frame(go_res), 15)

if (nrow(as.data.frame(go_res)) >= 5) {
  go_sim <- pairwise_termsim(go_res)
  options(repr.plot.width = 12, repr.plot.height = 10)
  print(emapplot(go_sim, showCategory = 20) +

ggtitle("Enrichment Map \u2014 GO Biological Process"))
} else {
  cat("Not enough GO terms for enrichment map.\n")
}

entrez_map <- bitr(
  all_de, fromType = "SYMBOL", toType = "ENTREZID", OrgDb = org.Mm.eg.db)
reactome_res <- enrichPathway(
  gene = entrez_map$ENTREZID, organism = "mouse", pAdjustMethod
  = "BH", pvalueCutoff = 0.1, readable = TRUE)
if (!is.null(reactome_res) && nrow(as.data.frame(reactome_res)) > 0) {
  cat("Top Reactome Pathways (Mouse):\n")
  print(head(as.data.frame(reactome_res)[, c("Description", "GeneRatio", "p.adjust", "Count")], 10))
  options(repr.plot.width = 10, repr.plot.height = 8)
  print(dotplot(reactome_res, showCategory = 15,
    title = "Reactome Pathway Enrichment \u2014 Mouse"))
} else {

```

```

    cat("No significant Reactome pathways found at padj < 0.1.\n")
}

sorted_res <- shrunk_res[order(shrunk_res$padj), ]
f_full <- "DESeq2_Ethanol_vs_Control_full_results.csv"
f_sig <- "DESeq2_Ethanol_significant_genes_padj01.csv"
write.csv(as.data.frame(sorted_res), file = f_full, row.names = TRUE)
sig_df <- as.data.frame(sorted_res[!is.na(sorted_res$padj) & sorted_res$padj < 0.1, ])
write.csv(sig_df, file = f_sig, row.names = TRUE)
cat("\u2713 Full results \u2192", f_full, "\n")
cat("\u2713 Significant genes \u2192", f_sig, "\n")
cat("Total significant (padj < 0.1):", nrow(sig_df), "\n")

```